Uncovering New Possibilities for Rice Agriculture: A CNN-SVM Model for Accurate Identification of Severity in Bacterial Brown Spot Rice Leaf's Disease

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Abstract— **This project aims to create a computer vision-based model for identifying and categorizing bacterial brown spot disease in rice leaves. If not recognized and treated right once, bacterial brown spots, a common condition of rice leaves, may drastically diminish crop production. However, many studies have yet to use computer vision techniques to detect and categorize this illness. The work fills this gap by suggesting a CNN-SVM-based model trained on 6432 photos of excellent and diseased rice leaves gathered from nurseries in Punjabi villages. The presented model achieved an accuracy of 99.826% and 95.247% in binary and multi-classification, respectively, and showed excellent accuracy in differentiating between four distinct degrees of bacterial brown spot disease severity. The model performed very well at detecting diseases with moderate severity.**

Further proving its superiority in identifying and categorizing this illness, the suggested model served better than other previously trained models in diagnosing bacterial brown spots in rice leaf disease. The creation of this model has significant ramifications for rice leaf bacterial brown spot disease management and prevention. Farmers can quickly take action to control the illness and avoid considerable crop losses by having a reliable and precise approach to identifying and diagnosing this disease. This research also lays the groundwork for creating hybrid disease detection and classification models that may be used to recognize other illnesses in rice leaves. Overall, this study advances the application of computer vision techniques for disease detection and classification in agriculture, which benefits sustainable rice production and food security.

Keywords—Convolutional Neural Network, Support Vector Machine, Deep Learning, Severity, Classification, Brown Spots.

I. INTRODUCTION

An estimated 140 million tonnes of rice are produced in India each year, making it one of the world's major producers. The northern and eastern parts of India are where most of the country's rice is grown; Punjab, West Bengal, Uttar Pradesh, and Andhra Pradesh are some of the top states for rice production. In India, rice is a staple food with a significant economic impact [1]. Additionally, the nation sends large amounts of rice abroad. One of the most important staple crops in the world, rice (Oryza sativa), provides more than one-fifth of the calories people eat daily. It is typically cultivated in tropical and subtropical areas, with Asia accounting for the bulk of output. Naturally, rice plants grow to 2 to 6 feet and have long, thin leaves. They are typically cultivated in paddy fields and flooded areas that provide optimal crop growth conditions. Low and highland regions are suitable for growing rice, and various kinds are adapted to different climates [2]. Rice leaves are green foliage that emerges from a rice plant's stem. They are generally long and thin and play a significant role in the plant's photosynthetic process. Rice leaves are usually between 10 and 40 cm long and 2 to 8 cm broad, typically green. They usually have a pointy tip and a tapering base and are placed alternately on the plant's stem.

Numerous animals, such as cows, goats, and elephants, eat rice leaves as a food source. Additionally, some farmers utilize the leaves as a source of organic fertilizer or to feed the leaves to their cattle. The fungus Cochliobolus miyabeanus causes brown spots, a widespread ailment of rice plants. If this illness is not addressed, it may result in considerable yield losses. Circular or asymmetrical dark or reddish-brown dots on the leaves, ranging in size from tiny to enormous, are one of the disease's symptoms. The spots, which often have a yellow halo around them, may cause the leaves to wilt and drop off. The fungus may affect any stage of the rice plant's development, but the reproductive phases are when it does the most harm. Spores that are transported by the wind and rain splashes propagate the illness.

Farmers may use cultural measures, including crop rotation, seed preparation and sanitation, and fungicide application to reduce brown spots. Additionally, it's crucial to choose resistant rice cultivars and to stay away from growing rice in recently flooded or humidified areas. Several illnesses may harm rice leaves, but a few more prevalent ones are as follows: The bacterium Xanthomonas oryzae is responsible for bacterial leaf blight[3]. Water-soaked sores on the leaves that develop yellow and brown and may result in leaf death are symptoms. Magnaporthe oryzae, a fungus, is what produced the blast. Circular or asymmetrical lesions on the leaves are one of the symptoms, which may cause a considerable reduction in output. Rhizoctonia solani is a fungus that causes sheath blight [4]. Brown or black sores on the leaf sheaths are one of the symptoms, and the plant may lodge or topple over as a result. Cochliobolus miyabeanus is a fungus that causes brown spots. The symptoms include circular or asymmetrical brown or reddish-brown dots on the leaves encircled by a yellow halo. Leaf folder is an insect-caused disease that manifests as folded or rolled leaves, which may interfere with photosynthesis and diminish production.

Utilizing resistant cultivars, crop rotation, treating seeds, maintaining good hygiene, and using fungicides and

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insecticides as necessary are all examples of preventive methods. For the early discovery of the damaged plants, prompt treatment, and, most importantly, developing plans to avoid illnesses to reduce losses, monitoring the diseases, their occurrences, and frequencies is crucial. Crop disease management in Bangladesh has historically included manual identification of any plant abnormality, expert assessment of that irregularity as a disease, and the recommendation of the right course of action. When vast farms are considered, this set of responsibilities becomes quite tricky [5]. Additionally, it adds more time and labour. The kind of illness, the susceptibility of the rice variety, and climatic circumstances are only a few of the variables that might affect how severe rice leaf diseases are [6]. Certain conditions, such as bacterial leaf blight and blast, may significantly reduce yield, but other diseases may have less effect when left unchecked.

Depending on the illness and the crop, there are several methods to gauge how severe a disease is. Examples include:

- The proportion of the disease's impacted leaf area.
- The number of infected tillers.
- The number of affected panicles.
- The number of afflicted grains per panicle.
- The disease's effects on grain production and quality.

For instance, if brown spot is severe and uncontrolled, it may result in a drop of up to 50% in grain output, while severe infestations of leaf folders can result in losses of up to 30% in yield. Farmers must periodically inspect their fields for disease indications to reduce crop loss and take necessary measures to manage them.

II. Literature Review

Based on a thorough review of India's second-largest rice producer. The research that has been done in this area is described in this section. The CNN method [7] has drawn many researchers' attention. The recognition accuracy of CNN is excellent. The researchers put forward a plan of attack to deal with the rice disease issue. Images of rice illness were collected to form an infield dataset. It was composed to classify rice illness. As a result, the experimental work has a 95% accuracy rate. The extent of the disease may be determined in large part by its severity, which also has an impact on crop output. The capacity to swiftly and effectively identify disease severity would aid in preventing production deficiencies; historically, educated experts identified disease severity by physically examining rice tissues [8]. According to the researchers, it gets protein and calories from rice as a dietary source [9].

Moreover, as the population grows, rice demand and consumption increase. By 2030, rice output must expand by more than 40% to meet rising food demand. Unfortunately, rice illnesses have significantly reduced yields. Rice blast disease is one of the primary offenders3, causing yields to drop by 60% to 100%4. The primary techniques for treating the illness are using pesticides and adopting cultivars resistant to blasts5. However, excessive pesticide usage drives up the price of rice production and seriously harms the environment6.

Furthermore, even for highly qualified professionals, diagnosing rice blasts is common in practice and is subjective and time-consuming. With the support of specialists in plant protection, to first provide a dataset on the rice blast disease [10]. To model is trained and validated using the dataset to identify agricultural conditions. Second, to offer a practical CNN-based feature extraction and classification approach for rice blasts, with classification accuracy levels exceeding 95% [11]. The infections must be identified to manage them and lessen the effects of the assaults. Unfortunately, in certain areas, there are sometimes no professionals accessible who can recognize the disorders. Observing the visible changes (disease spots or lesions) brought on by chemical changes in ill plants is one method of diagnosing plant illnesses. These spots' photos may be analyzed and utilized to identify diseases swiftly and affordably. This technique is likewise non-destructive, and the outcomes are reliable. To detect four prevalent paddy diseases in Indonesia with a minimum accuracy of 83.00%. If paired with additional characteristics, this strategy can be utilized as one of the features, particularly when two illnesses of about the same hue are involved [4]. The researchers show knowledge and use of new technologies, notably image analysis and, more recently, hyperspectral images, have made significant strides in the last thirty years. Both of these will become more important in determining the severity of an illness. Some typical causes of mistakes in visual rating have been found, and the science developing ideas of illness assessment. The strengths and weaknesses of methodologies used to evaluate visual disorders have started to be known empirically, and inter-heterogeneity exists. Comparing estimated and actual values is necessary to recognize and quantify error features. In comparison to the original data, an issue has been found and may be planned to be fixed. Therefore, creating a baseline of understanding mistakes throughout the process is crucial to identify the causes and measuring progress. Over the last 30 years, several investigations have shown the diversity of abilities among professional and novice visual assessors in various route systems that display a broad range of illness signs [12]. Because of its widespread use as a primary food product, rice is necessary to feed the world's expanding population. Early identification is essential because rice leaf diseases are dangerous to rice production. Rice illness diagnosis by conventional physical methods is time-consuming, costly, and useless. To solve this problem, a pre-trained algorithm that examines pictures of the diseased plant regions can recognize and categorize rice illnesses. Deep learning Convolutional Neural Networks (CNNs) effectively pull picture characteristics to identify diseases [13].

Regarding forecasting rice illnesses, this research seeks to reduce waste and handle time. The CNN model's suggested Improved Activation and Optimizer Function (IAOF) increases categorization and forecast success while decreasing loss. The research has proven that the IAOF-CNN model is the finest application and outperforms all other techniques [14].

III. Methodology

Since there hasn't been much use of earlier studies, identifying and classifying a bacterial brown spot rice leaf disease has been challenging. The suggested work is influenced by past research. However, the bacterial brown spot rice leaf disease hasn't been done yet. This methodology will help to understand and correctly complete the results mentioned above. Four independent procedures have been devised to accomplish the suggested task. The created

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diagrammatic representation approach, detailed in the next section of the study, is shown in Figure 1.

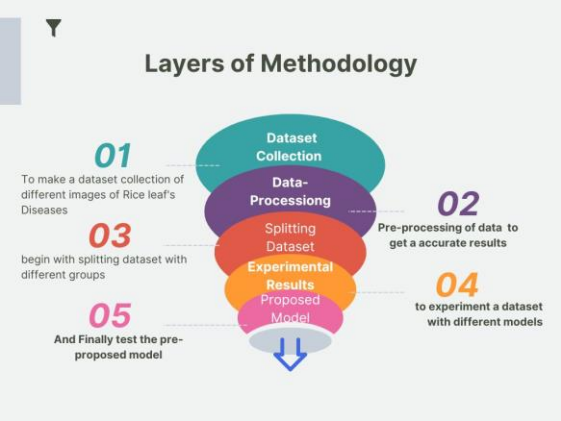
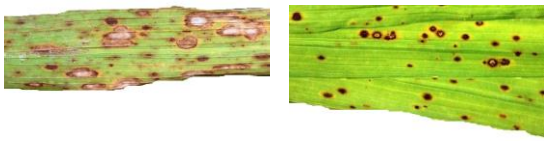


Fig. 1. Layers of methodology

A. Dataset Collection

Define The proposed study relies on building a DL model to recognize and classify a bacterial brown spot illness of rice leaves; as a result, the first stage in creating a model is to gather an image dataset. 6432 actual phase photos of rice leaves have been collected in a nursery in India to train and verify a model. The dataset's pictures of the brown spot were taken in a setting to create the model.

Images of brown spots on rice leaves were captured using a Canon DF600 camera. 6432 images were captured at different angles, ratios, and resolutions. Almost all photos captured have a size of 4284*2496 pixels and an 8–9 megabyte file size (MB). These guidelines are not needed to be followed since the captured photographs will be pre-processed in the next phase of the method. Before beginning the approach's pre-processing stage, every captured photograph was meticulously cross-checked to guarantee that the dataset was error-free. The photos were shot close to the brown spot rice leaf to retain picture quality. For example, images of the brown spots and healthy rice leaves are shown in Figures 2A and 2B.



2(a) 2(b)

Fig. 2. Diseases of Brown Spots Rice Leafs

B. Data-Processing

- Use uniformizing the collected data in size, resolution, and pixel brightness. A method based on geometric transformation used picture rotation and scaling techniques to resize the complete dataset collected depending on size.
- Avoid Most of the photos taken were 1080 × 1080 pixels. To ensure an appropriate image could be used as input, they were all reduced in size to 224 by 224 pixels and rotated at proper angles. Equation.
- The recommended pre-processing step must be used for results to be what is wanted.
- Preventing data under and overfitting at this pre-processing step also helps with performance. To pre-process all 6432 collected photos, which took close to a week to finish.

C. Experimental Results

As previously stated, all prerequisites for a successful experiment are included and shown by the experimental setup technique—6432 bacterial brown spot photographs, including healthy and pre-processed. To make a collection from the whole dataset suitable for further processing, since the pictures will be fed into the DL model for the classification of the bacterial brown spot leaf disease sickness, the entire dataset is rescaled, rotated, and brightened. Pictures were split into groups at a ratio of 85:15 after the pre-processing step.

One of the primary objectives of the experiment is to categorize images according to the severity of the bacterial brown spot rice leaf disease and to evaluate the effectiveness of the models for classifying images into binary and multiple classes. As mentioned above, the objective was accomplished by suggesting and using a DL model that could identify and categorize photos of bacterial brown spot rice leaf disease according to the severity of the illness. A model based on CNN has been created using the Keras framework and an SVM model in the Python programming language to categorize the dataset's picture shape. Convolutional classification makes up the whole model. A dataset is segmented based on these four classifications of bacterial brown spots in rice leaf disease severity. Table 1 provides examples of the four severity levels and their averages.

The severity ranges of bacterial brown spots on rice leaves are explained in the table supplied. The severity levels are divided into several groups based on the proportion of the leaf surface that the illness has impacted. The descriptions of each class are as follows:

- 1-25% (A) This class depicts the first phase of infection when the bacterial brown spot has just recently begun to harm the rice leaf. Only 1-25% of the leaf surface is impacted.
- 26-50% (X) This classification denotes a moderate stage of the illness. The infection has advanced by this time, and around 26–50% of the leaf surface is infected.
- The disease has progressed in this stage, and a significant section of the leaf is diseased. 51-75% (Y) Late moderate stage. The leaf's surface exhibits bacterial brown spot symptoms on around 51–75% of it.
- 76-100% (Z) Late level: This class corresponds to the disease's most advanced stage. At this point, a sizable leaf area is covered with the illness, which has spread widely. Bacterial brown spot affects 76–100% of the leaf surface.

It is simpler to evaluate the intensity and development of the illness in rice plants by classifying the severity levels into distinct groups. Researchers, farmers, and industry professionals can successfully monitor and control the

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growth of bacterial brown spots using these courses, as shown in Table 1.

TABLE 1. BACTERIAL BROWN SPOTS IN RICE LEAF'S DISEASE SEVERITY LEVELS

Bacterial Brown Spots in Rice leaf severity level	Explanation of Different Classes
1-25%	**(A)Initial stage:** The rice leaf infection caused by the bacterial brown spot has only begun.
26-50%	**(X)Moderate stage:** In this moderate stage of the illness, 27–52% of the leaf surface is affected.
51-75%	**(Y)Late moderate stage:** Rice disease with bacterial brown spot leaf has infected approx. 78% of the leaf at this stage.
76-100%	**(Z)Late level:** With a range of 79–100%, this disease stage covers most of the leaf.

For the experiment's smooth operation, build a straightforward CNN-based DL architecture using Python, the Keras framework, and the picture of 224 × 224 dimensions. It was verified using a variety of CNN model layers, primarily made up of sigmoid in the case of binary images with rice brown spots leaf disease and healthy pictures in the case of multi-classification.

The first layer in the CNN model is the convolutional layer. It applies size and other characteristics filters to the picture dataset. The photos are more accurately filtered using the Max-pooling layer's most pronounced characteristics. The dense layer then classifies the picture using the convolutional layer's output.

D. Hybrid Deep Learning of the Proposed Model

a. CNN-SVM Model

Convoluted, ReLU (activated), flattened, dense, max-pooling layers that make up the architecture are used to extract the main components of the input image. It is built with the use of convolution kernels. The kernel's settings are also altered per the picture's size. It illustrates how combining the two techniques may significantly improve the accuracy of disease identification for images. Figure 3 displays the architecture of generated model as in:

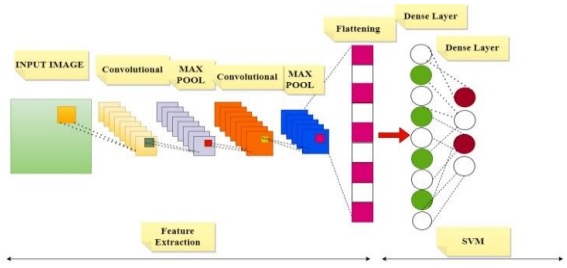


Fig3. CNN-SVM Model Architecture

To provide a CNN approach for both feature extraction and multiple picture classification. Ratio 85:15 has partitioned to dataset into two main halves, with 15 as the validated sample and 85 as the train set. The information is further separated into eight groups, one containing photographs of healthy brown spots in rice leaf disease, while the other seven provide images of ill leaves shown in Figure 4.



Fig4. An example of pre-processed Brown Spot Rice Leaves with various illnesses.

It is designed to picture multi-classification based on the four bacterial brown spots leaf disease severity levels on rice crop leaves. Two convolutional layers, max-pooling layers, dense layers, and one flattening layer comprise the seven layers of the CNN model. The total dataset was divided between training and testing halves at a ratio of 85 to 15 for this experiment, as mentioned in Table 2.

TABLE 2. PARAMETERS VALUES OF THE PROPOSED MODEL

Layer Type	Layer Name	Parameters
Input		Image size: 224x224x3
Convolutional	Conv1	Kernel size: 3x3, Filters: 32, Stride: 1, Activation: ReLU
Max Pooling	Pool1	Pool size: 2x2, Stride: 2
Convolutional	Conv2	Kernel size: 3x3, Filters: 64, Stride: 1, Activation: ReLU
Max Pooling	Pool2	Pool size: 2x2, Stride: 2
Convolutional	Conv3	Kernel size: 3x3, Filters: 128, Stride: 1, Activation: ReLU
Max Pooling	Pool3	Pool size: 2x2, Stride: 2
Convolutional	Conv4	Kernel size: 3x3, Filters: 256, Stride: 1, Activation: ReLU
Max Pooling	Pool4	Pool size: 2x2, Stride: 2
Flatten		Output size: 2,048
Dense	FC1	Output size: 512, Activation: ReLU
Dropout	Dropout1	Dropout rate: 0.5
Dense	FC2	Output size: 256, Activation: ReLU
Dropout	Dropout2	Dropout rate: 0.5
SVM	SVM	C: 1, Gamma: 'auto', Kernel: 'rbf'

The CNN-SVM model has four convolutional layers with max pooling, as shown in the chart, followed by two completely connected layers with dropout regularization. An SVM classifier is used to classify the output of the final ultimately linked layer.

The 6432 pictures collection was first divided into training, validation, and testing groups for preparation. The pictures were then adjusted to have a zero mean and unit variation before being reduced to a set dimension of 224x224. Data enrichment methods like random horizontal rotation and clipping were also used to expand the training set and enhance the model's generalization capabilities.

The model's effectiveness was assessed using several measures, including accuracy, precision, memory, and F1-score. The precise numbers for these measures would vary depending on the experimental design and dataset

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used. The CNN-SVM model has suggested appears to be a hopeful method for precisely assessing the degree of bacterial brown spot disease in rice stalks. The model might be used in real-world situations to assist producers in spotting and controlling illness in their rice harvests, improving agricultural output and minimizing financial losses. The architecture is run via supervised learning. Eigenvalues are another term for planar surfaces. The data classification for Brown Spot and Rice Leaf's images uses several types of SVM. Support vector machines refer to the decision boundaries of positive and negative support vectors as plane surfaces (SVM).

IV. RESULTS

The CNN-SVM DL model is intended to identify and categorize brown spots in pictures according to the severity of rice leaf disease. The four rice leaf disease classes have been divided into 6432 picture datasets. Precision, recall, and F-1 scores were used to measure the detection accuracy and performance, which ensured and demonstrated the correct efficiency and effectiveness of the created CNN-SVM. To get the suggested model's optimum detection and classification accuracy results, tweaking the hyperparameters such as steps per epoch and confusion metrics are shown in Table 3.

TABLE 3. CONFUSION MATRIX AT BROWN SPOTS LEAVES.

1048	**12**	**15**	**18**	**19**
11	**1175**	**14**	**13**	**15**
14	**15**	**1247**	**13**	**16**
18	**11**	**12**	**1284**	**16**
21	**18**	**19**	**14**	**1374**

In Table 4, a mention for parameter's metrics of the brown spots on rice leaves, a detection accuracy of 99.5%, has been noted. Additionally, the rice leaf disease severity with the late moderate level of brown spots has the most outstanding detection accuracy of 99.5%.

TABLE 4. PARAMETERS METRICS

Macro Average	95.24	95.25	95.25
Weighted Average	95.27	95.27	95.27
Micro Average	95.27	95.27	95.27

Additionally, Table 5 displays the performance of several parameters above the precision that resulted for each severity level of brown spots in rice leaf disease. It shows the error matrix that resulted from the classification of multiple classes of brown spots in rice leaf disease images.

TABLE 5. MEASUREMENT OF THE IMPLEMENTED MODEL'S PERFORMANCE

	Precision	**Recall**	**F1-Score**	**Support**	**Support Proportion**	**Accuracy**
Class 1	94.24	94.24	94.24	1112	0.17	0.98
Class 2	95.45	95.68	95.57	1228	0.19	0.98
Class 3	95.41	95.56	95.48	1305	0.20	0.98
Class 4	95.68	95.75	95.71	1341	0.21	0.98
Class 5	95.42	95.02	95.22	1446	0.22	0.98

The CNN model based on the DL methodology is used to build a method for identifying bacterial brown spots in rice leaf diseases. The experiment uses accurate data from the bacterial brown site in rice leaf diseases in an image dataset that includes pictures of the healthy and sick. It was run two times to obtain binary and multi-classification results of images based on four different levels of the bacterial brown spot severity in rice leaf disease photos in Figure 5 is mentioned below.

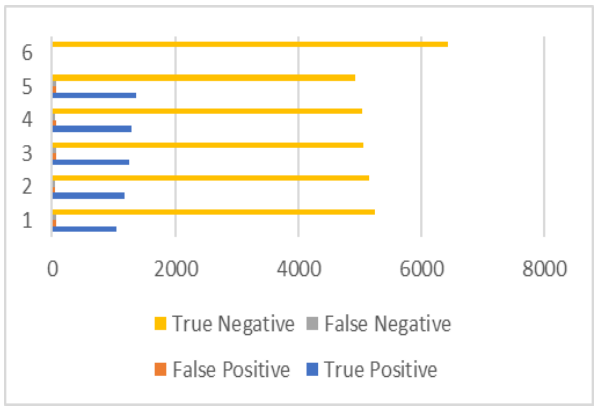


Fig5. Comparison of different parameters metrics

The accuracy of binary classification was 99.813%, and the best multi-classification accuracy was produced by the V severity level of the rice crop dataset at 95.274%. The proposed work's long-term objective is to increase the dataset with more diverse disease severity classification levels. Additionally, hybrid DL model implementation for a bacterial brown spot in rice leaf disease detection will be a significant future worry to models. The same is used for additional dataset pictures in the future.

V. CONCLUSION

Using 6432 images gathered from nurseries in Punjabi villages, the model has been presented to discriminate bacterial brown spots in rice leaf disease based on four distinct degrees of disease severity. A CNN-SVM was created to perform the bacterial brown area in rice leaf disease. The entire experiment was carried out using a three-step methodology. This methodology resulted in an accuracy of 99.826 % and 95.247% classes of the bacterial brown spot in rice leaf disease. By moderate severity of the bacterial brown area in rice, leaf disease has been shown to have the most incredible classification accuracy. The suggested model has also done better in classification when compared. With the creation of hybrid illness detection and classification models, the same model may be used for additional dataset pictures in the future.

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